

Simulation Model Structure Module (SMMM)

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Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

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Governed Space: Simulation Model Structure

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Canonical Language: English (UCL™)

Governed Space

Simulation Model Structure

Canonical Definition

Simulation Model Structure Module (SMMM) defines the governance framework for identifying, constructing, decomposing, organizing, documenting, and controlling the structural composition of a Simulation Model so that its components, relationships, dependencies, interfaces, internal states, interaction pathways, and structural boundaries remain explicit, traceable, and consistent with the governed Simulation Object, Purpose & Scope, Reality Representation, and Simulation Assumptions.

SMMM establishes the structural foundation through which a simulation representation becomes an identifiable executable model rather than an opaque technical implementation.

Operational Role

SMMM is the first internal governance space of the Simulation Model & Environment Standard (SMES).

It answers:

What is the Simulation Model structurally composed of, how are its components related, and what structural arrangement actually produces the behavior being simulated?

The governed progression is:

Governed Reality Representation → Governed Simulation Assumptions → Model Structure Definition → Component Identification → Relationship & Dependency Mapping → Interface Definition → Structural Boundary Establishment → Governed Simulation Model Structure

SMMM establishes the model structure upon which the remaining SMES governance spaces can operate.

Module Operational Space

SMMM governs: Simulation Model identification, model structural definition, model decomposition, model composition, component identification, component classification, component roles, component relationships, structural dependencies, model hierarchy, model topology, interaction pathways, internal interfaces, external model interfaces, state relationships, structural constraints, model coupling, model composition rules, structural abstraction, structural completeness, structural consistency, structural change visibility, model-structure version binding, representation-to-model traceability, assumption-to-model traceability, and structural reconstruction.

Module Operational Scope

SMMM applies to any Simulation Model whose internal structure materially affects simulation behavior, interpretation, reproducibility, comparability, or downstream reliance.

The module may govern: mathematical models, computational models, physics-based models, statistical models, behavioral models, agent-based models, AI-based simulation models, autonomous-system models, digital twins, system-of-systems models, discrete-event models, continuous models, hybrid models, multi-model simulations, coupled models, scientific models, industrial models, infrastructure models, economic models, biological models, and future simulation-model structures.

The required governance depth should remain proportional to the Simulation Purpose, Scope, complexity, and consequence.

Module Function

SMMM exists to prevent the Simulation Model from becoming a technical black box whose structural composition cannot be connected back to the simulation inquiry it is intended to represent.

Its function is to ensure that material model structure remains sufficiently explicit to determine:

what components exist,

what each component represents,

how components interact,

which components depend upon others,

where structural boundaries exist,

which interfaces connect the structure,

and:

how the complete model structure relates to the governed representation of reality.

Simulation Model Structure

A Simulation Model Structure is the governed arrangement of model components and the relationships through which those components collectively represent and execute the modeled behavior of the Simulation Object or relevant represented reality.

The structure may be:

Monolithic

Modular

Hierarchical

Distributed

Networked

Layered

Coupled

Composite

Agent-Based

or:

Hybrid

provided that its material structural properties remain governable.

Model Structure Identification

Every governed Simulation Model should have a sufficiently identifiable structural representation.

At minimum, where materially relevant, the structure should establish: model identity, model version, structural architecture, primary components, component relationships, material dependencies, interfaces, state relationships, structural boundaries, and relevant external dependencies.

A model name alone does not establish model structure.

Model Decomposition

Complex Simulation Models may require decomposition into governable components.

Conceptually:

Simulation Model

↓

Subsystem

↓

Component

↓

Subcomponent

↓

Relevant Structural Element

Decomposition should continue only to the level necessary for the intended simulation governance purpose.

SMMM does not require unnecessary decomposition where additional structural detail would not materially improve governance.

Model Component

A Model Component is an identifiable structural element performing a distinguishable function within the Simulation Model.

A component may represent: an object, actor, subsystem, process, rule system, behavior, physical mechanism, computational mechanism, AI model, controller, population, resource, network, or another modeled element.

Component Identity

Material components should remain distinguishable from one another.

Where two components perform materially different simulation functions, their identities should not be collapsed merely because they share the same technical implementation.

Component Role

Each material component should have an identifiable role within the model.

A Component Role may establish: what the component represents, what behavior it produces, which inputs it receives, which outputs it generates, which states it maintains, and which other components depend upon it.

Component Relationship

SMMM governs material relationships between model components.

Relationships may include:

Input → Processing

Cause → Effect

Parent → Child

Controller → Controlled Component

Producer → Consumer

Agent → Agent

System → Subsystem

State → Transition

or other simulation-specific relationships.

Structural Dependency

A Structural Dependency exists where the function, state, behavior, or interpretation of one model component materially depends upon another model component or structural element.

Dependencies should remain visible where their failure or modification could materially affect simulation behavior.

Dependency Direction

Where relevant, dependency direction should be identifiable.

For example:

Component B depends upon Component A

does not necessarily mean:

Component A depends upon Component B.

Bidirectional dependencies should be represented explicitly where they exist.

Model Hierarchy

Where a model contains hierarchical relationships, SMMM should preserve the hierarchy sufficiently to understand how lower-level behavior contributes to higher-level simulation behavior.

Conceptually:

System Model

↓

Subsystem Model

↓

Component Model

↓

Behavioral Element

Hierarchy should support governance, not merely documentation.

Model Topology

Where interaction structure materially affects simulation behavior, the topology of the model should remain identifiable.

This may include: networks, graphs, communication pathways, resource pathways, causal pathways, control pathways, or spatial relationships.

Interaction Pathways

A Simulation Model may produce behavior through sequences of interactions.

Material pathways should be traceable where necessary.

For example:

Input → Sensor Model → Interpretation Model → Decision Model → Action Model →
Environment Interaction

If a material pathway cannot be reconstructed, simulation behavior may become difficult to explain or compare.

Internal Interfaces

An Internal Model Interface connects two or more components within the Simulation Model.

Where materially relevant, an interface should establish: connected components, exchanged information or state, direction, format or representation, timing conditions, interaction rules, and applicable constraints.

External Model Interfaces

A model may also interact with elements outside its structural boundary.

External interfaces may connect the model to: Simulation Environment, input sources, external models, live systems, humans, AI systems, hardware, external services, or other simulation environments.

SMMM identifies the structural interface.

Governance of the broader simulation boundary interaction remains within the appropriate SMES governance space.

State Structure

Where a model contains states, the structural relationship between those states should remain sufficiently defined.

Examples:

Operational → Degraded → Failed

Idle → Active → Terminated

Untrained → Trained → Adapted

State transitions may materially alter model behavior.

Structural Constraints

Some model relationships are subject to structural constraints.

Examples may include: prohibited interactions, maximum connectivity, required component presence, dependency ordering, synchronization conditions, or topology restrictions.

Material structural constraints should remain explicit.

Model Coupling

Where multiple models or components influence one another, their coupling should be identifiable.

Coupling may be:

Loose

Tight

Unidirectional

Bidirectional

Synchronous

Asynchronous

Static

or:

Dynamic

where such distinctions materially affect simulation behavior.

Composite Simulation Models

A simulation may combine several models into a composite structure.

Conceptually:

Model A + Model B + Model C → Composite Simulation Model

SMMM requires the composition to preserve sufficient visibility into:
constituent models, their identities, versions, roles, interfaces,
dependencies, and interaction logic.

Composition must not erase material differences between constituent models.

Model-of-Models Structure

Some simulations contain models that themselves govern or invoke other models.

Where this occurs, SMMM should preserve:

Model → Submodel → Invoked Model → External Model

relationships sufficiently for structural reconstruction.

Structural Abstraction

Not every real-world element requires a separate model component.

SMMM permits structural abstraction where consistent with the governed Reality Representation.

However:

Structural simplification must not silently remove a material element that the governed Reality Representation requires the simulation to represent.

Representation-to-Structure Binding

The model structure should remain connected to the Reality Representation from which it was constructed.

Conceptually:

Reality Element → Represented Element → Model Component

This enables later determination of where a relevant aspect of reality appears within the executable model.

Assumption-to-Structure Binding

Material Simulation Assumptions may affect: component existence, component behavior, component relationships, dependencies, interfaces, state transitions, or structural constraints.

Where material, the relationship should remain traceable:

Simulation Assumption → Structural Element

An assumption should not become invisible merely because it has been encoded into model architecture.

Structural Completeness

Structural Completeness asks:

Does the governed model structure contain the material elements required by the approved Reality Representation and Simulation Purpose & Scope?

Completeness is relative to the governed simulation inquiry.

It does not require complete representation of reality.

Structural Consistency

Structural Consistency asks whether the components and relationships of the model remain mutually compatible with the intended structural design.

Examples of structural inconsistency may include: incompatible interfaces, contradictory dependencies, missing required components, invalid component relationships, incompatible state transitions, or broken composition rules.

Structural Coherence

A structurally complete model may still be incoherent if its elements cannot operate together as intended.

SMMM therefore distinguishes:

Structural Presence

from:

Structural Coherence.

No Hidden Component Principle

A material model component must not influence simulation behavior while remaining outside the governed model structure merely because it is technically embedded, externally invoked, or automatically generated.

No Hidden Dependency Principle

A material dependency must not be treated as irrelevant merely because it exists below the visible simulation interface.

No Structural Substitution Principle

A materially different structural implementation should not be treated as the same governed Simulation Model without assessing whether the change affects: representation, assumptions, dependencies, interfaces, states, execution behavior, or downstream interpretation.

Model Structure Change

A Model Structure Change may include: component addition, component removal, component replacement, hierarchy change, topology change, dependency change, interface change, state-model change, coupling change, or structural constraint change.

Material changes should remain identifiable.

Structural Change Classification

A change may be classified as:

Non-Material

Material

Critical

or:

Undetermined

relative to the Simulation Purpose & Scope.

Structural Change Propagation

A structural change may affect other SIMULOS® governance spaces.

For example:

Component Replacement

may affect:

Assumptions → Parameters → Inputs → Scenarios → Execution → Outputs

SMMM identifies the structural change.

Affected downstream spaces govern their respective consequences.

Model Structure Version Binding

The governed structure should remain connected to the applicable Simulation Model Version.

A simulation execution should therefore be capable of identifying which governed structural version was active.

Structural Reconstruction

SMMM should enable reconstruction sufficient to answer:

What was the model structurally composed of at the relevant time?

Which components existed?

How were they connected?

Which dependencies were active?

Which interfaces existed?

Which structural assumptions applied?

Which model version contained this structure?

Structural Traceability

The complete structural traceability path may be represented as:

Simulation Object

↓

Simulation Purpose & Scope

↓

Reality Representation

↓

Simulation Assumptions

↓

Simulation Model



Model Components



Relationships & Dependencies



Interfaces



Structural Boundaries



Model Structure Version

Minimum Implementation Framework

1. Identify the Simulation Model

Establish the model identity and applicable model version. 2. Define the Structural Architecture

Determine the material organization of the model. 3. Identify Material Components

Establish the components required for the governed simulation inquiry. 4. Map Relationships and Dependencies

Identify how material components interact and depend upon one another. 5. Establish Interfaces

Document material internal and external structural interfaces. 6. Bind Structure to Representation and Assumptions

Preserve traceability from governed representation and assumptions into the model structure. 7. Establish Structural Boundaries

Determine which elements belong to the model structure and which remain external. 8. Assess Structural Completeness and Consistency

Determine whether required structural elements are present and mutually coherent. 9. Bind Structure to Model Version

Ensure the governed structure can be associated with the correct model version. 10. Preserve Structural Reconstruction

Maintain sufficient records to reconstruct the relevant model structure later.

Governance Outputs

SMMM may produce: Simulation Model Structure Record, Model Component Register, Model Component Identity Record, Component Role Map, Model Hierarchy Map, Model Topology Record, Component Relationship Map, Structural Dependency Map, Interaction Pathway Map, Internal Interface Record, External Model Interface Reference, State Structure Record, Structural Constraint Record, Model Coupling Record, Composite Model Structure Record, Reality Representation-to-Model Map, Assumption-to-Structure Map, Structural Completeness Assessment, Structural Consistency Assessment, Structural Coherence Assessment, Model Structure Change Record, Structural Change Classification, Structural Change Propagation Record, Model Structure Version Binding, Structural Reconstruction Record, and Simulation Model Structure Traceability Map.

Use Case 1 — Autonomous Vehicle Simulation

A vehicle simulation contains separate models for:

Perception → Planning → Control → Vehicle Dynamics

SMMM identifies each as a material model component and governs their interfaces and dependencies.

If the planning component is replaced by a new AI model, the change cannot remain hidden merely because the rest of the simulation remains unchanged.

The new structural relationship and affected dependencies become visible for downstream governance. Use Case 2 — Multi-Agent Financial Market Simulation

A financial simulation contains thousands of autonomous trading agents interacting through a simulated market.

SMMM governs the structural classes of agents, market mechanism, interaction topology, information pathways, and dependencies required to understand how collective market behavior emerges.

The module does not determine whether the simulated market accurately predicts the real market.

It establishes whether the model structure producing the simulated behavior is identifiable and governable. Use Case 3 — Spacecraft Mission Simulation

A spacecraft simulation combines:

Flight Dynamics Model + Guidance Model + Propulsion Model + Thermal Model + Communication Model

SMMM preserves the identity, role, dependencies, interfaces, and version relationships of the constituent models.

If the propulsion model changes, the simulation can identify which structural pathways and dependent components may be affected before subsequent simulation execution.

Architectural Position

Simulation Model Structure is the first internal governance space of the Simulation Model & Environment Standard (SMES).

The complete SMES progression is:

Simulation Model Structure → Simulation Environment Configuration → Simulation Boundary Interaction → Simulation Parameterization → Simulation Environment Versioning

SMMM establishes the structural model basis required by the remaining SMES modules.

Within the broader SIMULOS® lifecycle:

Simulation Object → Simulation Purpose & Scope → Reality Representation → Simulation Assumptions → Simulation Model & Environment → downstream simulation governance

SMMM therefore governs the first internal transition from the conceptual representation of the simulation into the controlled executable structure through which simulation behavior will later be produced.

Foundational Principle

A simulation model cannot be structurally governed when the components, relationships, dependencies, interfaces, and assumptions producing its behavior cannot be identified and reconstructed.

Canonical Closing Statement

Simulation Model Structure Module establishes the structural governance foundation of the Simulation Model & Environment Standard by making the internal composition of a Simulation Model explicit, bounded, traceable, and reconstructable. Through component identification, decomposition, relationship and dependency mapping, hierarchy, topology, interface governance, state structure, coupling, structural abstraction, representation and assumption binding, change visibility, version binding, completeness, consistency, and reconstruction, SMMM ensures that simulation behavior cannot emerge from an invisible model structure whose relationship to the governed Simulation Object, Reality Representation, and Simulation Assumptions cannot be demonstrated.

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